
nginx_passwd

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NGINX-PASSWD

1.1 Overview

Basic Auth Password Manager. Manages basic auth password files. This replaces the functionality provided by ht-passwd from Apache. It also provides modern hash functions, such as argon2 and pbkdf2_sha512, which are far superior.

- All git tags are signed with arch@sapience.com key which is available via WKD or download from <https://www.sapience.com/tech>. Add the key to your package builder gpg keyring. The key is included in the Arch package and the source= line with *?signed* at the end can be used to verify the git tag. You can also manually verify the signature

1.2 New Or Interesting

Version 3.0.0

- Reduce our reliance on passlib (it is unmaintained).
 - passlib bcrypt is broken
 - there is a fork which works
- Algo changes no longer using passlib:
 - bcrypt now uses python-bcrypt directly
 - argon2 now uses python-argon2-cffi (which uses C library)
 - pbkdf2_sha512 now uses python-cryptography
 - pbkdf2_sha256 now uses python-cryptography

Older

- PEP-8, PEP-257, PEP-484 and PEP 561
- Refactor code
- Add pytests
- add python-bcrypt dependency to Arch PKBUILD
- For simplicity, make argon2 required instead of optional

GETTING STARTED

2.1 nginx_passwd application

2.1.1 Usage

To add or modify a user and write the resulting password file:

```
nginx-passwd -f <password_file> <user>
```

If file is not specified, then the result is written to stdout.

The supported algorithms are:

```
* Modern : argon2, pbkdf2_sha512, pbkdf2_sha256
* Active  : sha512, sha256, bcrypt
* Deprecated: md5, apr_md5/apr1
```

and sha256 is the default.

These older and now deprecated algorithms (*md5*, *apr_md5*) are still supported but should be replaced by one of the active ones.

Aside, *apr_md5* also known as *apr1*, is the ancient apache variant of md5.

2.1.2 Options

The options are given below and *nginx-passwd -h* provides a help summary.

Positional Argument:

- username
required argument.
- (*-h*, *-help*)
show help message and exit
- (*-f*, *-passwd_file*) <password_file>
Write to this Password file
- (*-a*, *-algo*) <algorithm>
Default is sha256. Can be one of:

```
* Modern : argon2, pbkdf2_sha512, pbkdf2_sha256
* Active  : sha512, sha256, bcrypt
* Deprecated: md5, md5_apr1
```

- *(-p, -passwd) <password>*

Password as an option. Without this it will be read from stdin.

- *(-D, -delete)*

Delete this user from the password file.

- *(-v, -verify)*

Checks that the provided password matches that in the password file

3.1 Installation

Available on

- [Github](#)
- [Archlinux AUR](#)

On Arch you can build using the PKGBUILD provided in packaging directory or from the AUR package. To build manually, clone the repo and do:

Listing 1: Manual Install

```
rm -f dist/*  
/usr/bin/python -m build --wheel --no-isolation  
root_dest="/"   
./scripts/do-install $root_dest
```

When running as non-root then set root_dest a user writable directory

3.2 Dependencies

- Run Time :
 - python (3.9 or later)
 - passlib

NB versions 1.1 and earlier used openssl - all newer version now use python passlib library.

- Building Package:
 - git
 - hatch (aka python-hatch)
 - wheel (aka python-wheel)
 - build (aka python-build)
 - installer (aka python-installer)
 - rsync
- Optional for building docs:
 - sphinx

- texlive-latexextra (archlinux packaguing of texlive tools)

3.3 Philosophy

We follow the *live at head commit* philosophy as recommended by Google's Abseil team¹. This means we recommend using the latest commit on git master branch.

3.4 License

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¹ <https://abseil.io/about/philosophy#upgrade-support>

LICENSE

nginx software is a Basic Auth Password File Manager for nginx.

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